Phenotypic impact of genomic structural variation: insights from and for human disease

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Abstract | Genomic structural variants have long been implicated in phenotypic diversity and human disease, but dissecting the mechanisms by which they exert their functional impact has proven elusive. Recently however, developments in high-throughput DNA sequencing and chromosomal engineering technology have facilitated the analysis of structural variants in human populations and model systems in unprecedented detail. In this Review, we describe how structural variants can affect molecular and cellular processes, leading to complex organismal phenotypes, including human disease. We further present advances in delineating disease-causing elements that are affected by structural variants, and we discuss future directions for research on the functional consequences of structural variants.

Microarray-based comparative genome hybridization
A technique that allows the detection of copy-number changes. It involves hybridizing differentially labelled test sample DNAs and reference DNAs to microarray slides with genomic probes, and measuring the intensity difference between the two samples.

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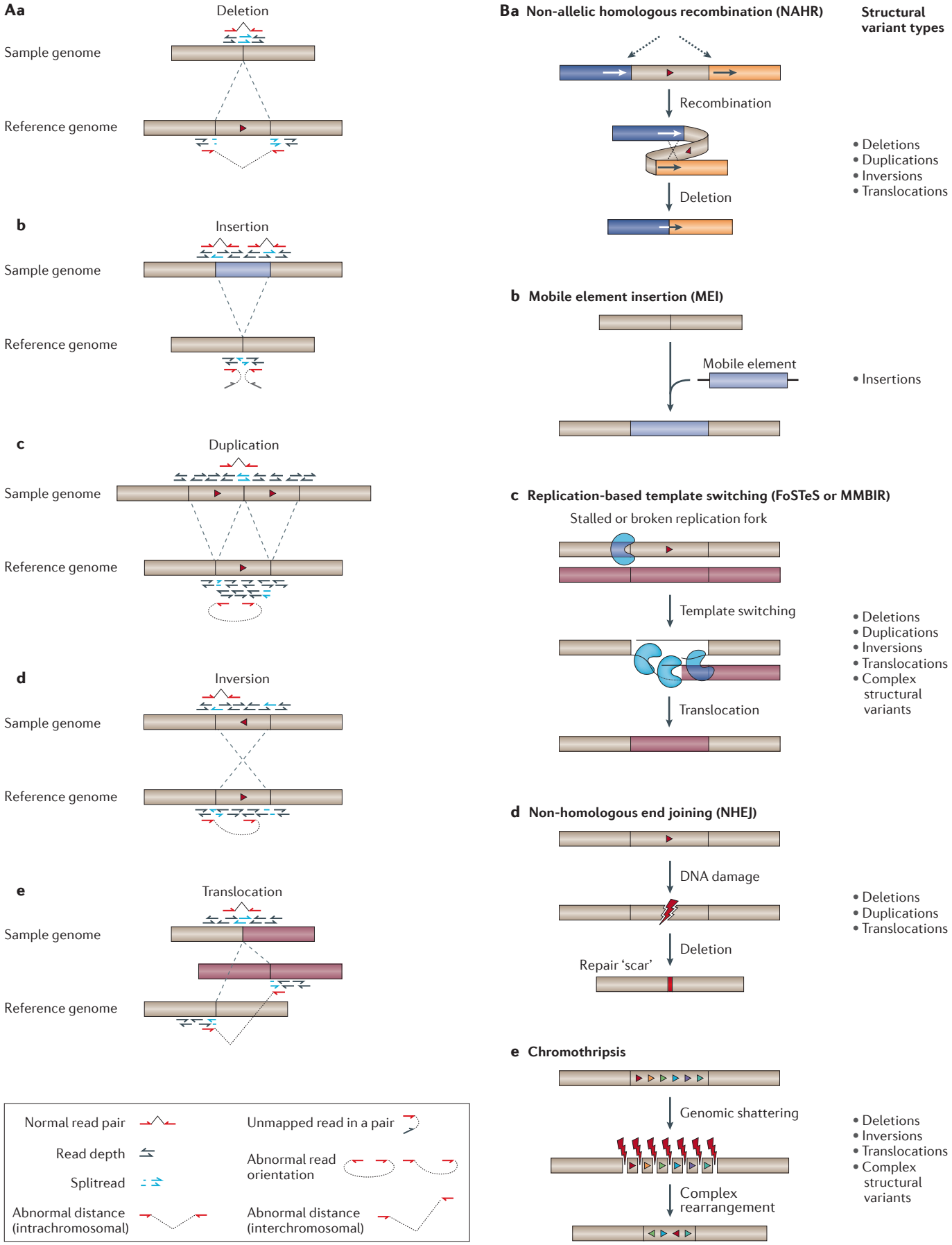
Among the forms of genetic variation that are present in the genome, structural variants have remained one of the most difficult to interpret with respect to their functional consequences. Structural variants comprise unbalanced forms of variation (leading to copy-number variation), including deletions, duplications, and insertions, and balanced forms, such as inversions and translocations[1]. They emerge as a consequence of different formation mechanisms including DNA recombination, replication and repair-associated processes (reviewed in REF. 2) (FIG. 1). A current operational definition, which is used throughout this article, classifies structural variants as sequence variants of at least 50 base pairs (bp) in size[3], thus discriminating them from smaller variants, such as single-nucleotide variants (SNVs) and short insertions and deletions (indels). Owing to their comparatively large size, structural variants are responsible for more heritable nucleotide sequence differences between individuals than SNVs (0.5–1% for structural variants and 0.1% for SNVs)[4,5]. Furthermore, structural variants can have relatively high locus-specific formation rates (*de novo* structural variants occurring at the same genomic locus among individuals) of up to 1 out of 7,000 newborns[6].

Chromosomal aneuploidies have long been known to be the cause of the well-known Down and Turner syndromes, and many submicroscopic large genomic rearrangements have been linked to various phenotypes[1,7] (TABLE 1). Recently, advances in structural variant detection technologies, including microarray-based comparative genome hybridization (array CGH) and high-throughput paired-end DNA sequencing techniques (reviewed in REF. 8) (FIG. 1) have greatly facilitated the ability to construct fine-resolution maps of structural variants in human populations and model organisms. Large-scale community efforts, including the 1000 Genomes Project[9] and disease genome sequencing studies (for example, the International Cancer Genome Consortium) have begun to generate nucleotide-resolution genetic variation maps and have used these to enumerate the extent and complexity of heritable[3,10] and somatically acquired[11] structural variants in hundreds of individuals. These maps have made it possible to associate structural variants with numerous common[4,12–15] and rare[16,17] human diseases, and there is mounting evidence that structural variants contribute substantially to the aetiology of these diseases, although some variants may have no phenotypic consequences[18].

Yet, molecular and mechanistic links between structural variants and phenotypes remain difficult to ascertain. The large size of structural variants, and the complexity of rearrangement-induced sequence alterations, can obstruct the interpretation of their effects. Furthermore, structural variants can impinge on functional elements in many ways, both inside of and beyond the genomic interval they occur in (FIG. 2).

In this Review we summarize lessons from current studies showing how genomic structural variants can

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◀ Figure 1 | **Structural variants: classes and formation mechanisms. A** | Structural variants comprise unbalanced copy-number variations ≥50 bp, including deletions (**a**), insertions (**b**) and duplications (**c**), as well as balanced variants such as inversions (**d**) and translocations (**e**). Widely applied DNA-sequencing-based approaches for structural variant detection using the relative orientation, position and read depth of paired-end DNA sequencing reads are indicated. For simplicity, read depth is represented by unpaired reads (reviewed in REF. 8). **B** | Molecular mechanisms leading to structural variant formation[145]. **a** | Recurrent structural variants often result from non-allelic homologous recombination (NAHR) which involves recombination between long highly similar low-copy-number repeats (blue and orange segments). **b** | Novel genomic insertions can involve mobile element insertion of transposable elements by retrotransposition. **c** | DNA-replication-associated template-switching events, involving the fork-stalling and template switching (FoSTeS)[146] and microhomology-mediated break-induced replication (MMBIR)[149] mechanisms, can lead to simple or complex structural variants, frequently involving duplicative events. **d** | Many structural variants in humans are attributed to non-homologous end joining (NHEJ), which is a process that repairs DNA double-strand breaks in the absence of extensive sequence homology and is often accompanied by the addition or deletion of several nucleotides in the form of a 'repair-scar' (small red bar). **e** | Chromothripsis — which is a phenomenon that seems to involve chromosome shattering leading to numerous breakpoints, followed by error-prone DNA repair — has been proposed to lead to one-off catastrophic rearrangements in several cancers and also in the context of germline DNA rearrangements.

result in various phenotypes and diseases, with a focus on heritable structural variants in mammalian genomes. We examine what can be learned from human disease phenotypes caused by structural variants, and discuss current efforts to obtain mechanistic insight into how these variants have an impact on molecular, cellular and organismal processes. We further highlight new results from animal models and computational modelling that are helping to explore the links between structural variants and phenotypes. Finally, we discuss the promise of novel technological and methodological advances that will aid the study of structural variants in a physiological context, and how findings from such studies may be used towards addressing translational questions in molecular medicine.

Structural variants in human disease

Major insights into the functional impact of structural variants have come from human disease studies. Several highly penetrant syndromes, involving microscopically visible or submicroscopic structural variants (TABLE 1), have played an important part in shaping our understanding of the phenotypic consequences of such variants and the diversity of the associated mechanisms. More recently, high-resolution genomic technologies have associated structural variants with complex, multifactorial disorders, and the insights gained from phenotypically well-studied Mendelian syndromes have proven invaluable for understanding their molecular aetiology.

Mendelian sporadic disorders involving structural variants. Sporadic disorders involving whole or partial chromosomal abnormalities that are detectable by traditional karyotyping (including chromosomal trisomies, monosomies and translocations, and segmental copy-number alterations of megabases of DNA) represent the most comprehensively studied disorders that are associated with structural variants. Prototypic examples include Smith–Magenis syndrome (SMS), Williams–Beuren syndrome (WBS) and Potocki–Lupski syndrome (PLS) (TABLE 1). Large low-copy-number repeats (also known as segmental duplications) frequently flank the respective variant region and make it prone to non-allelic homologous recombination (FIG. 1Ba). This results in recurrent DNA rearrangements (observed in an estimated 1/4,000–1/25,000 newborns[19,20]; see Online Mendelian Inheritance in Man (OMIM)) with almost identical breakpoint locations, which has led to the designation of such diseases as 'genomic disorders'[21]. The presence of these characteristic genomic sequence features has prompted the use of such large low-copy-number repeats to identify novel genomic loci that are susceptible to structural variant formation. Such a conceptual approach identified novel deletions in individuals with intellectual disability[22], and also identified duplications of regions that are known to be deleted in some syndromes, presumably as reciprocal events of these deletions[23–25].

Recurrent structural variants can lead to variable phenotypes, with distinct clinical manifestations. For example, partially overlapping deletions in the 22q11 band can lead to DiGeorge syndrome (DGS) and velocardiofacial (VCF) syndrome[26]. For some structural variants, a single gene seems to be responsible for most phenotypic effects, whereas in other cases, multiple genes contribute (TABLE 1). Phenotypes observed in disorders caused by megabase-size structural variants encompassing several gene loci (either protein-coding or non-protein-coding functional heritable units) are thought to largely involve alterations in gene dosage, rather than 'positional effects'. However, positional effects have also been documented in some instances, including in WBS, SMS and PLS models[27,28]. Of genetic disorders annotated in the Database of Chromosomal Imbalance and Phenotype in Humans using Ensembl Resources (DECIPHER) disease database, only 20% are caused by duplications, whereas 80% are caused by deletions, and duplications in the DECIPHER database are, on average, associated with a less-severe phenotype grading[29]. This can be explained by more-pronounced dosage alterations in the context of deletions (which result in a 1:2 variant-to-normal gene dosage ratio) compared with duplications (which result in a 3:2 variant-to-normal gene dosage ratio). Also, deletions can unmask recessive alleles that are present in the remaining copy of the respective region[30,31], and can expose inactive imprinted genes[32]. Different clinical manifestations occur when such imprinted genes are affected, depending on the parental origin of the variant (TABLE 1). Furthermore, whereas several chromosome duplications (trisomies) in humans are viable (trisomy 21, 18, 13 and X), all monosomies aside from monosomy X are embryonically lethal.

Complex disease and structural variants. The impact of structural variants on human diseases has recently been re-emphasized by numerous studies associating common diseases with submicroscopic (1–200 kb) structural variants, with the respective phenotypic traits often linked to complex, multilocus (that is, multifactorial)

Paired-end DNA sequencing
A method whereby DNA is fragmented and sequenced from both ends. If the DNA fragments are size-selected, discrepancies in the distance and/or orientation of the two ends compared with a reference genome can reveal the presence of deletions, duplications, inversions and translocations.

Karyotyping
Determining the number, form and size of chromosomes in the nucleus of a cell. Chromosomes are visualised by photomicrographs of isolated and stained chromosomes in metaphase.

Genomic disorders
Diseases that are associated with specific genomic architectures that favour recurrent genomic rearrangements, leading to abnormal dosage or dysregulation of the affected genes.

Positional effects
The effects on expression of a gene or gene region when its genomic context is changed, such as by translocation to another chromosome.

Imprinted gene
A gene for which expression is determined by the parent that contributed it.

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Table 1 | **Overview of prototypic structural-variant-associated human diseases and traits***

Disease or phenotypic trait	Type of structural variant	Region	Size	Causative genes	Type of change
Down syndrome	Aneuploidy (triplication)	Chr21	>10 Mb	Multiple	Increased dosage
Smith–Magenis syndrome (SMS)	Del	Chr17p11.2	3.7 Mb	Multiple, including *RAI1*	Haploinsufficiency
Potocki–Lupski syndrome (PLS)	Dup		3.7 Mb		Increased dosage
Williams–Beuren syndrome (WBS)	Del	Chr7q11.23	1.5–1.8 Mb	Multiple, including *ELN* and *LIMK1*	Haploinsufficiency
22q11 deletion syndrome (which includes velo-cardio-facial (VCF) syndrome) and DiGeorge syndrome (DGS)	Del	Chr22q11	1.5–3.0 Mb	Predominantly *TBX1*, but modifying loci include *COMT* and *CRKL*	Haploinsufficiency
Thrombocytopenia-absent radius (TAR) syndrome	CNV	Chr1q21.1	~2 Mb	*RBM8A*	Mutation or gene dosage
Distal 1q21.1 deletion/ duplication syndromes				Multiple, including *HYDIN2*	Gene dosage
Angelman syndrome (AS)	Maternal del	Chr15q11–13	Variable, ~3 Mb	*UBE3A*	Loss of function (imprinted)
Prader–Willi syndrome (PWS)	Paternal del		~3 Mb	Multiple, including *SNRPN* and *NDN*	Loss of function (imprinted)

*A more comprehensive version of Table 1, including references and descriptions of the disease phenotypes, can be found as Supplementary information S1 (Table). *APP*, amyloid beta (A4) precursor protein; CNV, copy-number variant; *COMT*, catechol-*O*-methyltransferase; *CRKL*, v-crk sarcoma virus CT10 oncogene homologue (avian)-like; Del, deletion; Dup, duplication; *ELN*, elastin; *HYDIN*, HYDIN axonemal central pair apparatus protein; *LIMK1*, LIM-domain-containing protein kinase 1; *NDN*, necdin; *RAI1*, retinoic-acid-induced 1; *RBM8A*, RNA-binding motif protein 8A; *SNRPN*, small nuclear ribonucleoprotein N; *TBX1*, T-box 1; *UBE3A*, ubiquitin protein ligase E3A.

patterns of inheritance (TABLE 1). Owing to the multiple genetic and environmental contributing factors, the inference of genetic determinants for such common diseases is particularly challenging. Advances in genome-wide association studies (GWASs), and the construction of population reference sets for structural variants[3,4,33] (that is, genotype sets that can now be used by imputation in SNV-based GWASs), have helped to uncover many associations between structural variants and such disorders, including attention deficit hyperactivity disorder, Crohn's disease, rheumatoid arthritis, and type 1 and type 2 diabetes[15,34–37]. Furthermore, improvements in genomic technologies have enabled rare *de novo* structural variant formation events to be identified in the context of sporadically occurring disorders, including autism and schizophrenia[17,24,38–41] (TABLE 1).

Interestingly, structural variants associated with complex phenotypes frequently intersect with Mendelian disease loci. For example, two recent surveys identified associations of *de novo* duplications at the 7q11.23 locus with autism spectrum disorders (ASDs) and schizophrenia[24,25]. At this locus, deletions cause WBS, which is characterized by engaging, hypersocial personalities; this phenotype is quite 'opposite' to ASDs. Duplications associated with ASDs have also been described in genomic intervals involved in 22q11 deletion syndrome, and the Prader–Willi and Angelman syndromes[24] (TABLE 1). There have also been other reports of copy-number changes leading to 'opposite' phenotypes, such as microcephaly and macrocephaly, which are associated with deletions and duplications at 1q21.1, respectively[42]. Similarly, obesity is linked to deletions at 16p11.2, and the underweight phenotype is linked to duplications at 16p11.2 (REF. 43). Further lessons can also be learned from Mendelian disorders. For example, deletions and duplications of a gene can also result in the same disorder, such as proteolipid protein 1 (*PLP1*) deletions, duplications and triplications in the Pelizaeus–Merzbacher X-linked recessive hypomyelinative leukodystrophy[44]. Alternatively, they can give rise to partially overlapping phenotypes, as is seen with the peripheral myelin protein 22 (*PMP22*) gene. Copy-number alterations of this gene lead to demyelinating neuropathies, resulting in either hereditary neuropathy with liability to pressure palsy (HNPP) in the presence of deletions, or Charcot–Marie–Tooth disease type 1A in the presence of duplications[45]. Collectively, these findings point towards the existence of dosage-sensitive genes (BOX 1) in these regions, the perturbation of which is linked with disease phenotypes in a context-specific manner.

Challenges in interpreting structural variant disease phenotypes. Adding to the difficulty of their ascertainment, structural variants often display variable expressivity and penetrance, further discussion of which has been covered elsewhere[1,7]. Variable expressivity can even occur in individuals of the same family[24,38], complicating genetic counselling. For example, structural variants observed at 1q21.1 are associated with a broad spectrum of variable phenotypes, including microcephaly, a severe form of congenital heart malformation and more-complex phenotypes including ASDs, schizophrenia and developmental delay[38,42,46]. Environmental factors and/or other genetic variants probably contribute to variable expressivity and penetrance. For example, in patients with thrombocytopenia-absent radius (TAR) syndrome, microarray-based structural variant analyses revealed the pervasive presence of a 200 kb deletion on 1q21

Imputation
A technique whereby a genomic variant is not directly genotyped, but instead its presence or absence is inferred on the basis of its association (co-occurrence) with another, genotyped variant.

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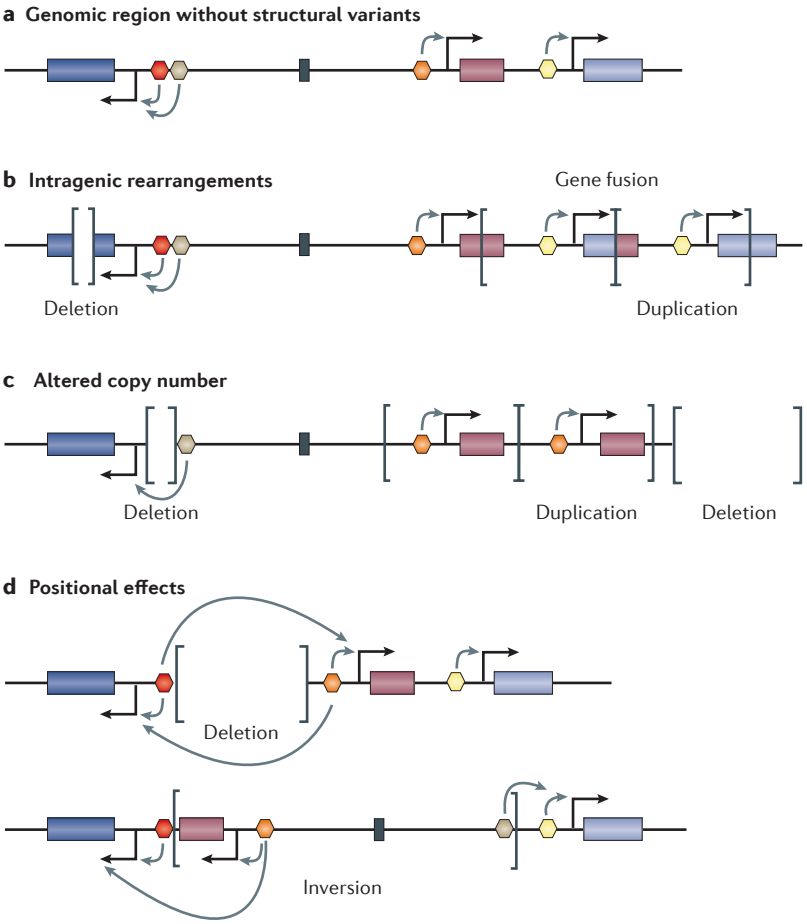


Figure 2 | **Functional consequences of structural variants. a** | Genes (boxes) are regulated by the collective and combinatorial input of regulatory elements, including tissue-specific enhancers (hexagons, with different colours indicating tissue-specificity, and arrows pointing to the target gene) and insulators (black rectangles), which block the activity of regulatory elements. **b**–**d** | Structural variants (shown by square brackets) can have phenotypic consequences by altering coding regions. For example, they can remove part of a coding region or fuse different coding regions after a duplication, resulting in aberrant transcripts (**b**). Alternatively, deletions or duplications can lead to altered doses of otherwise functionally intact elements (**c**), resulting in altered regulatory input (left) or altered gene copy number (right). Structural variants can also affect the expression of genes outside of the variants (that is, a positional effect[55,64]) (**d**), thus resulting in a gain or loss of regulatory inputs.

encompassing 11 genes[47]. However, the observed autosomal-recessive inheritance of the disease, and the presence of this otherwise rare deletion in unaffected parents of TAR patients, suggested the existence of another causative allele. Indeed, by exome-capture-based high-throughput DNA sequencing in TAR patients, a recent study identified a low-frequency SNP in the 5′-untranslated region and first intron of the remaining copy of *RBM8A*, which is one of the genes that is present in the deleted region[48]. Functional studies showed that the low-frequency SNP is associated with reduced expression of *RBM8A*, implying that TAR syndrome is caused by compound inheritance of a low-frequency hypo-active regulatory variant of *RBM8A* in addition to a deletion of the gene-containing region[48] (TABLE 1).

Epistatic interactions
Modification of the expressivity of an allele by one or several other genes (known as 'modifier genes'). In Crohn's disease, epistatic interactions may explain up to 80% of the so-called 'missing heritability'.

cis-regulatory elements
Genomic regions (for example, enhancers) that regulate the expression of genes on the same chromosome.

Further to the presence of additional variants allelic to the regions affected by structural variants, epistatic interactions involving distinct loci can contribute to phenotypic variability, as has been observed in Bardet–Biedl syndrome[49] and proposed in Crohn's disease[50]. Collectively, these effects probably contribute to the so-called 'missing heritability' that is frequently described in the GWAS context[51]. Finally, it is noteworthy that different platforms can lead to the identification of largely different sets of candidate genes, as is the case for ASDs (reviewed in REF. 52); this interesting observation may be connected with differences in the sensitivity and/or specificity of the respective genomic platforms.

Molecular consequences of structural variants

The molecular consequences of structural variants are intuitively straightforward to deduce when the variant encompasses entire genes — the expression of which is assumed to vary accordingly — or when the breakpoints of structural variants intersect with coding regions, thus leading to gene fusions or truncations. But structural variants can also influence genes over large distances, leading to measurable effects on gene expression up to hundreds of kilobases away from the rearranged site (FIG. 2).

The impact of gene copy number on mRNA levels. Consequences of gene copy-number changes on mRNA expression levels have been studied comprehensively in fruitflies[53], mice[54–56] and transformed human cell lines[57,58]. These studies concluded that globally there is indeed an appreciable correlation between mRNA levels and gene copy number. Analyses of cells derived from patients with full-length or partial chromosomal aneuploidies, including Down syndrome[59], WBS[27], or using engineered mouse models of the SMS and PLS syndromes[28], yielded similar correlations. Notably, however, for individual genes mRNA levels often deviated from the expected levels: that is, they were not halved when one gene copy was deleted, nor increased by a 3:2 ratio in a trisomic state. Further to this, not all genes with altered copy number displayed altered expression, and a small proportion even showed expression changes that were inverse to the copy-number alteration[57,59].

Systematic analyses of nucleotide-resolution structural variant breakpoints revealed that some apparent expression discordances can be accounted for by inaccuracies in breakpoint assignments[57], and other observed discrepancies might be explained by complex biological principles. For example, regulatory mechanisms can buffer against copy-number changes through epistatic interactions or autoregulatory feedback mechanisms at the individual gene level (FIG. 3), or even through compensation mechanisms acting on larger regions, as illustrated by dosage compensation of the sex chromosomes[60]. In addition, structural variants that overlap *cis*-regulatory elements (FIG. 2d,3d) can be inversely correlated with the expected directional change of mRNA abundance. For example, deletions can lead to the upregulation of gene expression in cases in which the deletion affects a silencer or insulator element[27].

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Expression quantitative trait locus (eQTL)
A genomic locus that regulates the mRNA expression level of a gene.

Effect size
The percentage of phenotypic variance that can be attributed to a given genomic variant.

Interestingly, gene expression studies in patients with duplications, and expression quantitative trait locus (eQTL) mapping in mice, have additionally revealed that structural variants do not merely affect the levels of gene expression, but also its variability[55,59]. This suggests that regulatory feedback loops, with the role of maintaining expression levels at a constant level, may act imperfectly in the event of gross copy-number changes.

Further to this, an eQTL survey carried out in seventeen inbred mouse strains disclosed that structural variants display, on average, larger effect sizes than SNVs[61]. Although the measurable effects are typically largest when structural variants intersect with genes, many eQTLs (or phenotypic QTLs) are attributable to structural variants in intergenic regions[57,62,63], indicating that these structural variants may exert their effects by altering gene regulatory regions. Indeed, measurable effects of structural variants on expression can extend in *cis* far beyond the breakpoints of these variants[27,55,64], presumably because *cis*-regulatory elements are affected. Moreover, eQTLs are commonly identified as being cell-type-specific, tissue-specific and/or developmental-stage-specific[54,55,64], in keeping with gene expression programs operating in a highly context-dependent manner.

Box 1 | **The relevance of gene dosage**

Predicting the phenotypic consequences of structural variants remains difficult, partially owing to our incomplete knowledge of genes that are haploinsufficient (in which half a dose is detrimental) or dosage sensitive (in which both increased and decreased gene dose is detrimental). Screens in *Saccharomyces cerevisiae*[136], *Drosophila melanogaster*[137] and *Caenorhabditis elegans*[138] suggested that the proportion of such genes is relatively small under normal conditions (~3% of genes in yeast and flies), but considerably increases under nutrient-limited or stressed conditions (for example, to 12–20% of genes in yeast[139]). So far, no similar data exist for humans, but indirect evidence indicates that a certain tolerance to changes in gene dosage also occurs in humans. For example, the 1000 Genomes Project is sampling individuals from the 'normal' population, and it was recently estimated that in these individuals at least 24–28 genes are affected by large deletions[140]. It remains to be assessed to what extent these changes in gene dosage are truly tolerated (without phenotypic consequences), or have mildly deleterious phenotypic consequences. Recent surveys of published literature identified 146 human genes that are known to be dosage sensitive[141] and 299 haploinsufficient genes[142]. An obvious caveat is that complex phenotypes (such as disease susceptibility) are only revealed under extreme circumstances (for example, pathogen exposure), and are probably missed in such analyses. Additionally, gene dose may contribute to normal phenotypic variability, thus causing additional complexity.

Dosage compensation may serve as an explanation for the lack of extreme phenotypes at certain gene loci. Through dosage compensation, expression changes of dosage-altered genes are either adjusted to their normal levels ('buffered'), or their detrimental effects are masked through genetic interactions. Local and chromosome-wide dosage compensation has been documented for certain dosage-sensitive regions, such as imprinted loci or the mammalian X chromosome, and disruption of these compensating mechanisms has pathogenic consequences[32,143]. Although there is no experimental evidence for a generic genome-scale buffering of gene copy-number changes in humans, evidence for tissue-specific regulatory effects on individual genes has been presented[27,55]. Such regulation may occur through feedback loops to adjust mRNA or protein levels. In addition, phenotypic buffering may involve genes with redundant functions[140] or interconnected gene regulatory circuits[144], both of which have evolved to tolerate gene expression noise. Whether, and to what extent, each of these mechanisms contributes to the tolerance of altered gene dose remains to be investigated.

Structural variants and regulatory elements. Assessing the mechanism behind expression changes due to structural variants is challenging, in part, as the annotation of *cis*-regulatory elements in mammalian genomes is not nearly as complete as it is for genes. In addition, *cis*-regulatory systems are frequently composed of multiple individual modules that control distinct spatiotemporal aspects of gene expression, and that can be found at considerable distance (>100 kb) from promoters. Therefore, structural variants that affect only a discrete portion of the *cis*-regulatory landscape of a gene can have tissue-specific or developmental-stage-specific effects, which can phenocopy the tissue-specific loss-of-function of a gene[65]. β-thalassaemia, caused by deletion of the *β-globin* locus control region[66], and holoprosencephalies due to translocations separating the sonic hedgehog (*SHH*) gene from the regulatory elements controlling its expression in the developing forebrain[67] are prototypical examples of this. Similarly, Pierre Robin syndrome (PRS), which is a malformation of the palate due to misplacement of the tongue and micrognathia, was recently shown to be caused by distinct microdeletions — 3 Mb apart — affecting candidate *SOX9* developmental enhancers[68].

Apart from abrogating gene expression, structural variants can also induce overexpression or ectopic expression. For example, a precise level of bone morphogenetic protein (BMP) signalling is crucial to balance proliferation, apoptosis and differentiation during limb development[69]. Duplications encompassing a limb enhancer that controls *BMP2* can cause localized overexpression of this signalling molecule, leading to the development of hypoplastic digits[70]. The same logic (additional copies of tissue-specific enhancers leading to localized overexpression) could apply to large duplications flanking *SOX9* that are associated with Cooks syndrome or Indian hedgehog (*IHH*) that are associated with syndactyly and craniosynostosis[71,72].

The mechanisms leading to pre-axial polydactyly — when regions including the limb enhancer of *SHH* are duplicated — are probably more complex: this enhancer normally drives expression in the posterior region of the developing limb bud, but the phenotype implies that *SHH* is ectopically expressed in the anterior limb bud following duplication[73,74]. Importantly, similar ectopic expression has been observed due to point mutations in the limb enhancer itself, which are proposed to affect the recruitment of transcription factors to this enhancer[75]. Along the same lines, a recent study showed that the colorectal hereditary mixed polyposis syndrome (HMPS) is caused by a 40 kb duplication between gremlin 1 (*GREM1*) and secretogranin 5 (*SCG5*) leading to increased and ectopic expression of *GREM1* in the intestinal crypt epithelium[76]. Particularly large structural variants can also lead to ectopic expression owing to positional effects by connecting genes with regulatory regions that they are not usually associated with[77,78] (FIG. 2d). Together, these examples indicate that structural variants can alter the intrinsic activities of regulatory elements, possibly by revealing latent (but normally repressed) regulatory potential.

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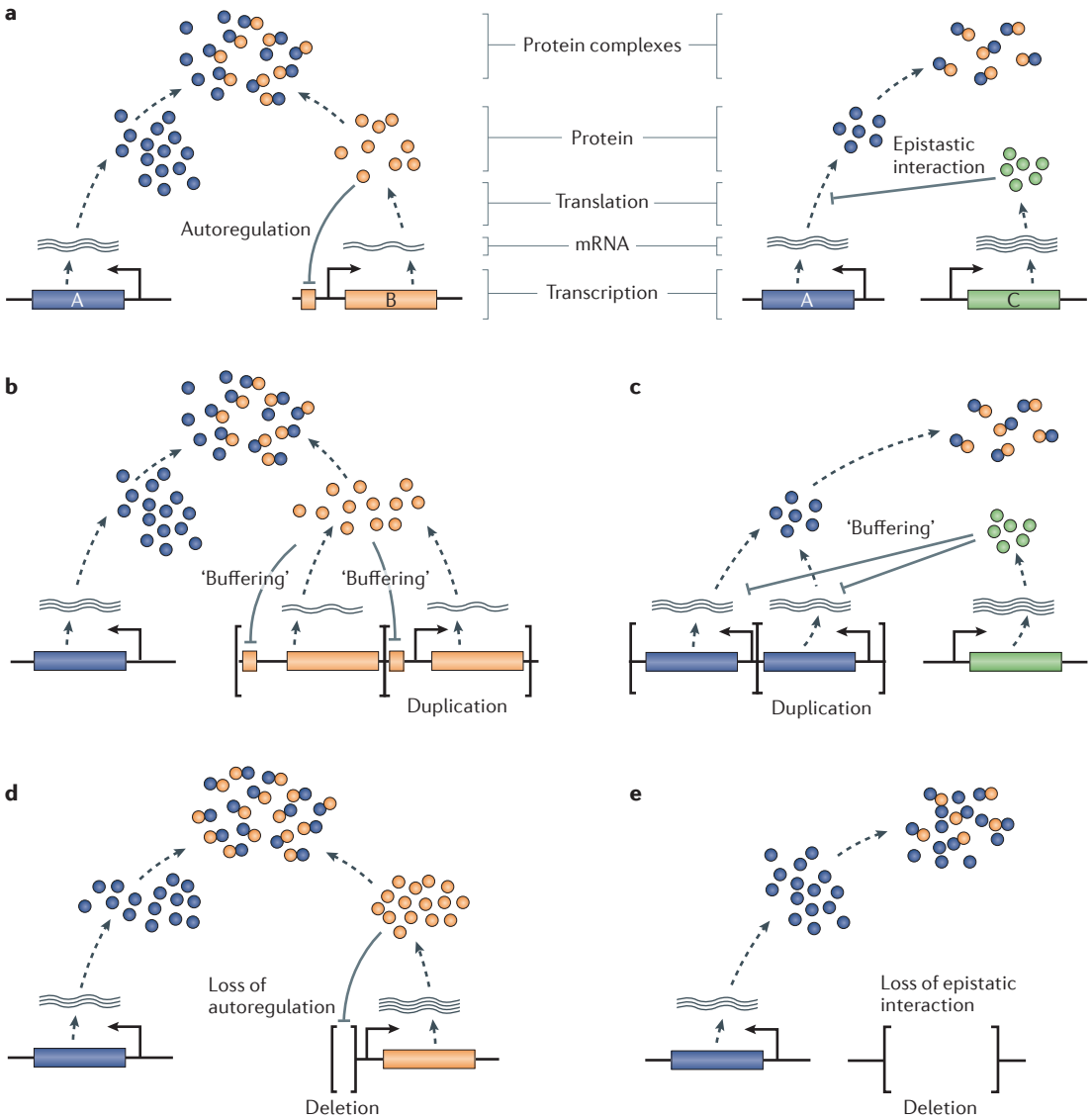


Figure 3 | **From genotype to phenotype: buffering and feedback.** **a** | Genetic information is converted into cellular phenotype through a multi-tiered process: genes A (blue), B (orange) and C (green) are transcribed into mRNA and translated into proteins, which then form higher-order protein complexes. This process is highly regulated at the level of transcription and translation, and through the control of mRNA and protein degradation rates. In addition, downstream products can affect any of these steps through regulatory feedback loops. Here, protein B regulates its own transcription rate through a *cis*-regulatory element. Interactions between gene products can modulate the resulting phenotypes, shown here by protein C regulating the translation rate of protein A, and protein A forming functional complexes with protein B. Accordingly, altered gene dosage as a result of structural variation can be buffered by various mechanisms. **b** | Autoregulation of the transcription of B by its protein buffers the increased abundance of protein B. **c** | Cross-regulation may occur whereby the regulation of translation of transcript A by protein C buffers the effect of gene duplication. **d** | Alternatively, the dosage of a limiting factor can be increased by loss of autoregulation, whereby protein B is no longer able to regulate its transcript production owing to loss of a regulatory element. **e** | Additionally, dosage of a factor can be increased through the loss of an epistatic regulator, such as the loss of gene C resulting in increased translation of protein A. Furthermore, dosage alterations influencing higher-order processes, such as signalling and phosphorylation, could have an impact on phenotype, as recently proposed for a mouse model of Down syndrome[147].

Structural variants and protein expression levels. Although most studies relating structural variants to expression changes have investigated mRNA levels, ascertaining how these relate to protein-level changes is crucial. Several recent studies have addressed this question by measuring correlations between mRNA and protein levels in *Caenorhabditis elegans*, *Drosophila melanogaster*[79] and different mammalian species[80,81]. These studies reported an overall modest correlation between mRNA and protein levels, particularly in mammals, in which genome-wide no more than 40% of protein abundance can be explained by mRNA abundance[80–82],

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although the proportion increases when translation rates[81] or factors influencing those rates[80] are taken into account. Hence, changes in mRNA levels caused by copy-number alterations do not necessarily result in corresponding changes in protein levels, as protein levels are additionally modulated by post-transcriptional regulation, translational control, protein folding and stability, and higher-order regulatory interactions between genes and proteins (FIG. 3). In keeping with this imperfect correlation, studies mapping QTLs that are responsible for protein abundance (pQTLs) have pinpointed many loci that are not observed by eQTL mapping[82].

Despite the imperfect correlation between mRNA levels and protein expression, results from eQTL surveys can facilitate the establishment of links between structural variants and cellular and organismal phenotypes. For example, eQTL mapping has linked duplications of glyoxalase 1 (*Glo1*) to mouse strain-specific anxiety behaviour[64], and structural-variant-associated eQTLs have also been successfully linked to metabolic traits in mice, including weight and cholesterol levels[56].

Linking structural variants and phenotypes

Bridging the knowledge gap between structural variants and phenotypes requires precise structural variant information and the identification of the elements that are responsible for the observed trait. This is particularly challenging for structural variants that intersect with many functional elements.

Ascertaining structural variant breakpoints and allelic status. Structural variants often occur in repeat-rich, recently segmentally duplicated genomic areas. Thus, mapping breakpoints can be complicated by the reduced signal-to-noise ratios of high-throughput genomics technologies[8]. Furthermore, structural variants can arise from complex alterations involving numerous breakpoints (FIG. 1). Catastrophic chromosome alterations, termed chromothripsis[11] (FIG. 1Be), in which tens to hundreds of breakpoints are thought to form in a single dramatic structural variant formation event, constitute an extreme example of such complexity. Although initially observed in several types of cancer[11,83], evidence for similar events in the germ line has been presented[84,85]. Delineating the fine-scale architecture of structural variants, including such complex rearrangements, is facilitated by progress in paired-end DNA-sequencing-based data analysis[11,83,86].

Beyond breakpoint precision, knowledge of the allelic status of structural variants is crucial. It has been estimated that 11% or more[4,87] of the loci harbouring common, heritable structural variants are multi-allelic; that is, the respective genomic locus underwent multiple independent or concurrent DNA rearrangement events. Difficulties in assigning locus copy numbers in recently duplicated genomic regions[8] have imposed considerable problems on dissecting the phenotypic contribution of such structural variants. Nonetheless, the functional relevance of repeated duplications has been elucidated in a few cases. For example, at the *AMY1* locus, between 2 and 10 copies of the gene encoding salivary amylase have been detected, which correlate proportionally with protein abundance and show an association with population-specific differences in starch consumption[88]. The recent development of computational approaches for paralogue-specific duplication genotyping using short-read-based high-throughput sequencing data (reviewed in REF. 8) should facilitate the inference of copy-number state-specific traits also in other regions of the genome.

Mapping minimal critical regions. Having ascertained the genomic regions that are affected by a structural variant, the remaining challenge is to identify the genes inside or around the structural variant that are of functional relevance. In the context of genomic disorders that are caused by rare, recurrent structural variants, mapping of such genomic rearrangements from patients with the same phenotypic manifestation, but with different, partially overlapping variants, has been used to pinpoint critical regions of minimal genomic overlap (see the references in Supplementary information S1 (table)). The value of minimal critical region mapping is generally maximized by the availability of patients with overlapping phenotypes and non-identical (that is, 'atypical') breakpoints. Using such a rationale helped to delineate genomic regions that are likely to be causative for phenotypes that are commonly observed in Down syndrome[89,90]. Owing to advances in high-throughput DNA sequencing it has recently become possible to screen large cohorts of patients to identify gene-disruptive rearrangements (in addition to point mutations) in order to pinpoint single disease-relevant candidate genes. Examples include the recent finding that haploinsufficiency of *KANSL1* (a gene involved in histone acetylation) is responsible for the manifestations of the 17q21.31 microdeletion syndrome[91], and the uncovering of several gene loci that are probably involved in neurodevelopmental phenotypes by assessing single-gene disruption events at nucleotide resolution[92].

However, the resolution of minimal critical region mapping depends on the availability of patients with overlapping phenotypes and atypical genomic rearrangements. Also, the approach has a higher chance of pointing to specific elements in the case of loss-of-function alleles, compared with gain-of-function situations such as the re-allocation of regulatory elements[77,78].

Animal models of structural variants. Whereas tissue-specific or developmental-stage-specific effects of structural variants are usually inaccessible through human biopsy samples, they can be studied *in vivo* using animal models, which enable researchers to assign the phenotypic effects of these variants to key functional elements (TABLE 2).

With the development of new genomic approaches it has become possible to catalogue naturally occurring structural variants in many mammalian species, and this has been exploited to study their influence on specific traits. Such an approach is not limited to classical model organisms (such as mice), and has been used, for example, to delineate structural variants that are involved in the phenotypic diversity of dog breeds[93] or that are

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Table 2 | **Animal models of human disease-causing structural variants***

Human structural variant	Type of model	Model‡	Comments
Down Syndrome	Radiation-induced	Ts65Dn mouse	Partial trisomy of Mmu.16 (syntenic to Hsa.21); cognitive, early postnatal lethality
	Engineered aneuploidies	Dp(16Lipi-Zfp295)1Yey and other mouse strains	Partial trisomy of Mmu.16 (syntenic to Hsa.21); large duplications could be rescued by small engineered deletions
	Engineered 'transchromosomic' mice	Tc(Hsa21)1TybEmcf mouse	Mice carrying a freely segregating, extra copy of an almost entire human chromosome 21
Smith–Magenis or Potocki–Lupski syndromes; deletion or duplication on 7q11.2	Engineered deletions or duplications	Series of mouse strains with overlapping deletions or duplications	Partial phenocopy of the corresponding human genomic condition; phenotype dependent on deletion size and genetic background, suggesting the presence of modifying genes; altered *Rai1* gene dose is sufficient to capture a subset of phenotypes
	Gene deletion and duplication	*Rai1*-knockout mice	
DiGeorge syndrome; velocardiofacial syndrome	Engineered deletions	Mice strains	Large deletions narrowed down the interval responsible for the heart defect; validated by targeted knockout of *Tbx1*
	Gene knockout	*Tbx1*-knockout mice	
Prader–Willi syndrome; Angelman syndrome	Transgene-associated deletion	Del(*7Herc2-Mkrn3*)13FRdni mice	Deletion alleles of whole region or imprinting centre recapitulate the parent-of-origin-dependent phenotypes
	Targeted deletion including *cis*-regulatory module	*Snrpn*^{tm1Cbr} mice	
	Gene knockout	*Ndn*-knockout mice	Single gene knockouts validate the function of these genes in establishing the human phenotypes
	Gene knockout	*Ube3a*-knockout mice	
Split hand foot malformation, type 3	Spontaneous, ERV-insertion	*Dactylaplasia* mouse strain	Identical phenotype (missing central digits in both hands and feet), but different types of structural variants
Mesomelic dysplasia, Kantaputra type	Radiation-induced	*Ulnaless* mouse strain	Impact on long-range regulatory enhancers leading to ectopic regulation of *Hoxd* genes in limbs, through a complex duplication (Kantaputra) or a balanced inversion (*Ulnaless*)
Dup(15q11-13)	Engineered aneuploidies	Dp(7Herc2-Mkrn3)1Taku mice	Autism-like behavioural phenotypes, but only in duplications of paternal origin
	Transgene, overexpression	Tg(*Ube3a*) mouse strains	Transgene not imprinted, autism-like behavioural phenotypes regardless of parental origin
Del/Dup(16p11.2)	Engineered aneuploidies	Del/Dup mouse models	Deletion and duplications of the syntenic intervals
	Knockdown or overexpression	Zebrafish	Overexpression or knockdown of *kctd13* in zebrafish mimics microcephaly phenotype found in human 16p11.2 del/dup patients, and is associated with dysregulated proliferation of neural progenitors

*A more comprehensive version of Table 2, including references, can be found as Supplementary information S2 (Table). ‡ Where applicable, we indicate the official name of the mouse model, or if multiple similar models exist, an overview of these models (names from the Mouse Genome Informatics website). Del, deletion; Dup, duplication; ERV, endogenous retrovirus; *Hoxd*, Homeobox D cluster; Hsa, refers to a human chromosome; *kctd13*, potassium channel tetramerization-domain-containing 13; Mmu, refers to a chromosome from the *Mus musculus* mouse; *Ndn*, necdin; *Rai1*, retinoic-acid-induced 1; *Snrpn*, small nuclear ribonucleoprotein N; *Tbx1*, T-box 1; *Ube3a*, ubiquitin protein ligase E3A.

associated with parasite resistance in cattle[94]. Model animals with genomic variants that map to regions that are orthologous to human structural variants can further help to narrow down causative elements or mechanisms. For example, in humans, 0.5 Mb microduplications on 10q24 comprising 2–6 genes are associated with dominant limb malformations[95,96]. These observations suggested that increased gene dosage directly due to this segmental trisomy could be responsible for the phenotype. However, in mice this condition is phenocopied by insertions of endogenous retroelements at distinct positions in the orthologous interval, without any duplication[97,98]. Thus, the presence of these inserted retroelements next to, but outside of, the duplicated interval of a gene controlling limb outgrowth, suggests instead that these duplications may cause the limb malformations through a long-range regulatory effect[98].

Mice are often the model of choice to study disease-causing structural variants, as they show both physiological similarity and extensive synteny with humans. Moreover, with isogenic mice, in controlled environments, it is possible to reduce the variability that is observed in human patients, and hence to ascertain even relatively small phenotypic effect sizes (see for example REFs 99,100 and references therein). Furthermore, the possibility to extensively re-engineer the mouse genome, from point mutations to entire chromosomes (reviewed in REF. 101), facilitates their wide applicability for phenotypic analysis (TABLE 2). For example, the contribution of individual genes to Down-syndrome-associated phenotypes has been studied using a large collection of mice with different duplications (and deletions) of regions that are syntenic to human chromosome 21 (reviewed in REF. 90). Engineered deletions and duplications of

Synteny
Shared genomic organization between related species. It is usually seen as a shared relative order of genes or other functional elements on a portion of a chromosome.

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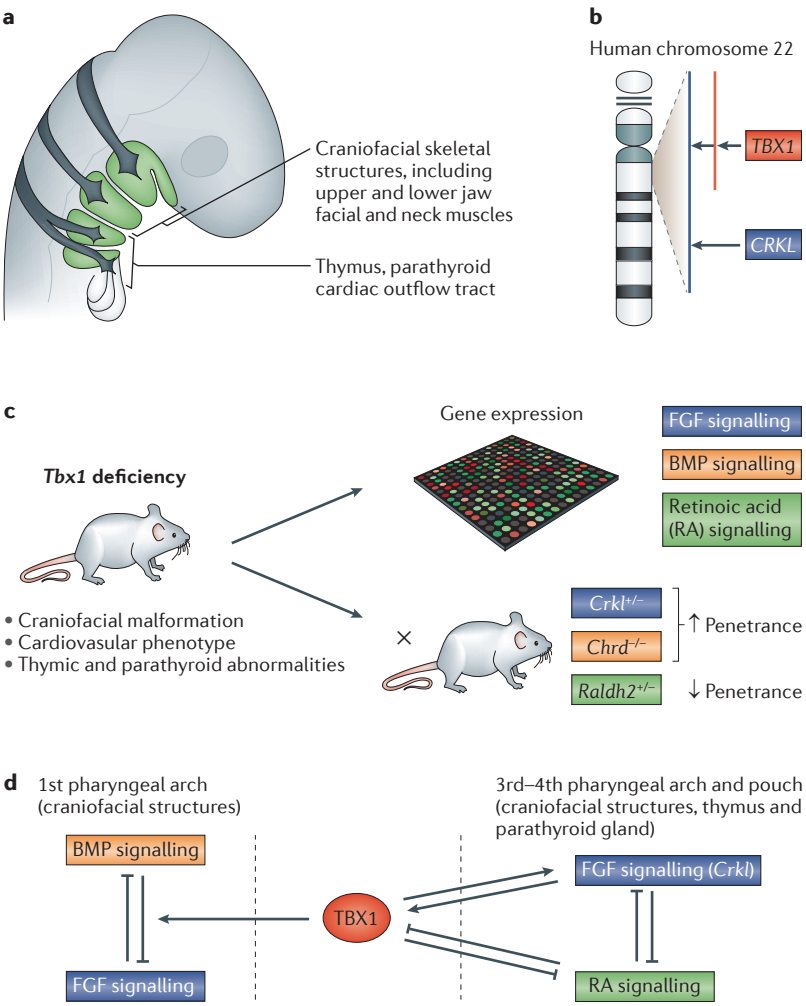


Figure 4 | **Reconstructing the genetic network in DiGeorge syndrome. a** | DiGeorge syndrome (DGS) is an autosomal dominant disorder that is characterized by congenital heart defects, thymic and parathyroid hypoplasia, typical craniofacial malformations and behavioural traits. All primary affected structures are derived from the embryonic pharyngeal apparatus (coloured in green) and the neural crest cells, which migrate into the pharyngeal pouches and cardiac outflow tract. **b** | Initially, analysis of patients with DGS with rare, microscopically visible chromosomal translocations revealed that the phenotype correlated with deletion of the chromosome 22q11 region[148]. Subsequent systematic analysis showed that DGS is typically caused by *de novo* recurrent deletions in that region (blue bar), which is mediated by flanking low-copy-number repeats[149,150]. This recurrent deletion contains *CRKL*, which is an adaptor protein for fibroblast growth factor (FGF) signalling. Most frequently, deletions are approximately 3 Mb in size (in 80% of cases), although a smaller deletion (red bar) results in almost equivalent symptoms. This smaller region contains *TBX1*, which is a transcription factor that is involved in pharyngeal arch development. **c** | As the paucity of additional atypical deletions in patients hampered minimal critical region mapping, the search for causative genes was carried out using mouse models. In mice, deficiency of *Tbx1* (through targeted deletion[103,104,151] or naturally occurring point mutations[113]) is sufficient to generate most DGS defects, although some traits show incomplete penetrance and weak expressivity. Gene expression analysis of mutant mice revealed multiple perturbed signalling pathways, including the FGF, bone morphogenetic protein (BMP) and retinoic acid pathways[109,110]. Subsequently, *Tbx1*-deficient mice were bred with mice in which key components of these pathways (indicated in the same colours as above) had been knocked out. This increased the penetrance of craniofacial malformations when the BMP pathway was disturbed[113], and of the thymic and cardiovascular phenotypes in FGF pathway mutants[105,111]. Also, a reduction in retinoic acid signalling reduced the penetrance of thymic malformations[111]. **d** | Collectively, these findings outline different signalling networks in the different pharyngeal arches (structures that subsequently develop into the structures mentioned in part **a**), which are involved in the DGS phenotypes.

the SMS interval demonstrated that most craniofacial, growth and behavioural abnormalities arise from gene dosage effects[102]. However, in this mouse model, the restoration of gene copy number was insufficient to rescue fully the abnormal neurobehavioral traits associated with the locus, thus revealing the concomitant existence of positional effects that impinge on gene expression[28].

An additional value of animal models is that they allow dissection of the genetic networks that contribute to different phenotypes. As an example (FIG. 4), engineered deletions using transgenic and knockout models have demonstrated that haploinsufficiency of the transcription factor *TBX1* is the major cause of the cardiovascular defects that are found in VCF and DGS[103], and that penetrance and expressivity are associated with *Tbx1* expression levels in a nonlinear manner[104]. Notably, mice heterozygous for a *Tbx1* null allele showed only partial penetrance and weak expressivity of the cardiovascular and pharyngeal arches defects, but these are enhanced in compound *Tbx1* and *Crkl* heterozygous animals[105]. *Crkl* encodes an adaptor protein that mediates fibroblast growth factor (FGF) signalling and maps adjacent to the minimal 1.5 Mb interval that is defined for 22q11 deletion syndrome and DGS, and is included in the larger 3 Mb recurrent deletion (FIG. 4). *Fgf8* mutations in mice lead to neural crest defects that resemble 22q11 deletion syndrome[106]. Further studies showed that TBX1 regulates *Fgf10* and *Fgf8* (REFs 107,108). Hence, 22q11 deletion syndrome is caused by reduced expression of different genes that act in the same TBX1–FGF pathway. Interestingly, the mouse models showed changes in retinoic acid signalling in the pharyngeal arches, including altered expression of genes that are involved in retinoic acid metabolism[109,110]. These elegant findings suggest that the identified genes and fetal exposure to vitamin A (from which retinoic acid is derived) could act as modifiers of the phenotypes that are associated with 22q11 deletion syndrome[111]. Also, analysis of *Tbx1*-mutant mice indicated that the gene is required to establish the mutually restrictive expression of *Fgf8* and *Bmp4* in the first pharyngeal arch[112]. In agreement with these findings, compound mutants that are deficient for *Chrd* (an antagonist of BMP signalling) and that show reduced *Tbx1* expression have increased penetrance of the craniofacial malformations[113]. *In vivo* models thus allow ascertainment of modifying factors, and could be used to identify and assess potential therapeutic targets.

Beyond mammalian systems, other model organisms in which gene expression levels can easily be manipulated are also increasingly used to delineate causal genes. In this regard, a recent study using systematic morpholino-mediated knockdown and mRNA overexpression in zebrafish demonstrated that the gene encoding the potassium channel component *KCTD13* is the major gene responsible for the reciprocal macrocephaly and microcephaly phenotypes that are associated with copy-number variation at the gene-rich 16p11.2 locus[114].

However, when making use of animal models it should be kept in mind that species- and strain-specific differences exist. As an example, mouse models of SMS

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phenocopy neither the typical heart and renal defects nor the ophthalmological abnormalities that are characteristic of SMS patients[102], and the craniofacial phenotype is influenced by the genetic background of the mouse strain used[115,116]. This highlights that dosage sensitivity can vary, not only due to different development timing or altered genomic context between species, but also because of the presence of different alleles of modifying genes within a given species[113,117,118]. Divergence in gene content (for example, due to species-specific gene duplications) and syntenic breaks can further complicate the generation and interpretation of animal models. For example, head size abnormalities that are associated with structural variants at 1q21.1 are proposed to result from the dosage alteration of a *HYDIN* gene paralogue, *HYDIN2*, which arose by interchromosomal duplication in recent primate evolution[42]. Yet, sophisticated chromosomal engineering strategies, such as targeted substitutions of large intervals[119], can help to construct mouse models with 'humanized' structural variants[120]. In this respect, the Down syndrome mouse model constructed by the Fisher and Tybulewicz groups[121], which has a freely segregating copy of human chromosome 21, constitutes not only an elegant tour de force, but also an excellent and useful model of this condition.

Computational approaches. The growing amount of information available from structural variant maps, gene expression data sets, and other large-scale data sets, makes computational modelling approaches an attractive complementary strategy for deciphering the functional impact of structural variants. Recently, advances have been made in two areas: first, in predicting the properties of disease-causing genes within structural variants; and second, in identifying networks and cellular processes that are disrupted in disease. Initial efforts to identify key genes that are affected by structural variants focused on determining the properties of haploinsufficient genes. Specifically, Huang and co-workers[122] found that such genes are typically longer and show a higher degree of evolutionary conservation than haplosufficient genes; haploinsufficient genes are also usually expressed at higher levels during early development, and exhibit more interaction partners in a probabilistic human functional interaction network[122]. Once established, these parameters could serve as useful predictors to distinguish disease-causing deletions from non-pathogenic ones. Using phenotypic information from mouse models, Webber and colleagues[123] recently developed a computational predictor of genes that are involved in intellectual disability and developmental delay. They searched for statistical associations between genes that are affected by rare structural variants of a patient cohort and phenotypic categories that are associated with the knockout mouse strains of those genes. The authors observed an enrichment of several candidate genes involved in diverse nervous system phenotypes in mice, thus suggesting that this approach could predict pathways and candidate genes involved in structural-variant-associated developmental delay[124] that could be prioritized for further analysis.

Recent studies have also focused on identifying cellular processes and networks that are perturbed in disease. This is particularly important for revealing the underlying characteristics of complex diseases, in which the individual contribution of genes is minor, but in which synergistic effects of networks amplify those effects. To this end, computational approaches have identified networks that are involved in synaptogenesis, axon guidance and related molecular processes within rare, *de novo* structural variants assessed by array CGH from patients with ASDs and intellectual disability[125,126]. Similarly, a recent association study, combined with a network analysis of recurrently affected pathways, reported cellular proliferation, projection and motility, GTPase–RAS signalling and genes in the ubiquitin pathway (*FBXO40*, *PARK2*, *RFWD2* and *UBE3A*) to be overrepresented in SNP-array-detected structural variants linked to ASDs[17,127].

Perspectives and future challenges

Although considerable improvements have recently been made in mapping structural variants and interpreting their functional impact, numerous important challenges remain. Present studies are mostly directed to 'unique' regions of the genome, but are 'blind' towards the phenotypic contribution of structural variants in complex, repeat-rich, highly duplicated areas of the genome, which are difficult, or even impossible, to ascertain using current genomics technologies. This will probably improve with future advances in DNA sequencing technology[8,128], including longer DNA reads to map variants in complex genomic areas.

Understanding the impact of structural variants on complex phenotypes will require more detailed and extensive data from large cohorts of patients, including improved phenotypic and clinical descriptions and associated genotype information, as well as tissue-specific expression data. Correct and comparable genotype–phenotype correlation and interpretation is highly dependent on sample quality (which includes standardization of the clinical and phenotypic information of these cohorts), sequencing data production and computational analysis. Large medical sequencing projects, such as Deciphering Developmental Disorders and UK10K, are specifically devoting efforts to collect such data. Furthermore, the International Standards for Cytogenomic Arrays (ISCA) Consortium has recently launched a whole-genome array database, which provides clinicians and researchers free access to a searchable catalogue of ranked disease-associated structural variants and affected genes, thus offering a fast and user-friendly interface to query a locus of interest. Other consortia, such as Finding of Rare Disease Genes in Canada (FORGE Canada) and international cancer genome projects are in the process of collecting high-quality cohorts in other contexts (for example, to study SNVs affecting Mendelian traits, and to study somatically acquired DNA alterations), which future studies can explore further for linking heritable or acquired structural variants to phenotypes. Within these disease studies, an increasing number are now using exome-capture-based genome sequencing to assess primarily SNVs

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and indels. Although such studies have the advantage of reduced cost and turnaround time compared with whole-genome sequencing, it is important to realize the highly limited resolution of capture-based approaches in terms of structural variant mapping, especially for small variants, variants in non-coding genomic areas and balanced variants such as translocations.

In addition to identifying structural variants, several technologies are allowing further investigation of these variants with regard to their role in phenotypes. For example, recent technological advances are improving our understanding of mammalian regulatory landscapes[129] and the three-dimensional organization of the genome[130]. This could potentially guide our understanding and predictions of long-range effects mediated by structural variants. Further to this, recent large-scale mapping of regulatory regions across many different cell lines by the Encyclopedia of DNA Elements (ENCODE) consortium now provides an unprecedented resource for scrutinizing the phenotypic effects of intergenic structural variants in a tissue-specific setting[131]. Furthermore, advances in *in vitro* technology — to generate disease-specific induced pluripotent stem cells (iPSCs)[132,133] — and advances in *in vivo* chromosome engineering strategies[129,134], are yielding suitable model systems for reconstructing pathogenic variants, including duplications, inversions and translocations. In addition, the development of site-directed nucleases is opening new avenues for generating targeted structural variants[135]. With these, the functional impact of structural variants can be probed more systematically, thus yielding an improved handle on the intricate structural variant–phenotype relationship. In addition, large-scale resources in model organisms, such as the International Knockout Mouse Consortium, provide growing catalogues of loss-of-function alleles to facilitate the analysis of individual candidate genes.

Altogether, the complexity of structural variant–phenotype relationships calls for novel integrative and holistic approaches that involve the combined application of multiple models for well-phenotyped patients. Such an integrative approach will facilitate the interpretation of the vast amounts of whole-genome data that are presently being generated in patients. This will form an essential prerequisite for improving predictions of disease onset and outcome, for developing therapeutic strategies and for facilitating translational molecular medicine research.

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Competing interests statement
The authors declare no competing financial interests.

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Database of Genomic variants: http://projects.tcag.ca/variation
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Deciphering Developmental Disorders: http://www.ddduk.org
Finding of Rare Disease Genes in Canada (FORGE Canada): http://www.genomebc.ca/portfolio/projects/health-projects/finding-of-rare-disease-genes-in-canada-forge-canada
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